CSE105/209A: Modern Algorithmic Toolbox,[*] PSET-2

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1.

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \tag{1}$$

subtract λ from diagonal

$$A - \lambda I = \begin{pmatrix} -\lambda & 1 & 0 & 0 \\ 1 & -\lambda & 1 & 0 \\ 0 & 1 & -\lambda & 1 \\ 0 & 0 & 1 & -\lambda \end{pmatrix} \tag{2}$$

the characteristic polynomial is

$$\lambda(\lambda(\lambda^2-1)-\lambda)+(-1(\lambda^2-1)) = \lambda^4-3\lambda^2+1 = (\lambda^2-\lambda-1)(\lambda^2+\lambda-1) \tag{3}$$

which gives roots of

$$\begin{aligned} \lambda_1 &= -\frac{1}{2} - \frac{\sqrt{5}}{2} \\ \lambda_2 &= \frac{1}{2} - \frac{\sqrt{5}}{2} \\ \lambda_3 &= -\frac{1}{2} + \frac{\sqrt{5}}{2} \\ \lambda_4 &= \frac{1}{2} + \frac{\sqrt{5}}{2} \end{aligned} \tag{4}$$

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to find eigenvalues, find the null space after subtracting λI from the diagonal of A.

$$(A-\lambda I)x = 0 = \begin{pmatrix} -\lambda & 1 & 0 & 0 \\ 1 & -\lambda & 1 & 0 \\ 0 & 1 & -\lambda & 1 \\ 0 & 0 & 1 & -\lambda \end{pmatrix} x = 0 \tag{5}$$

$$\begin{aligned} -\lambda x_1 + x_2 &= 0 \\ x_1 - \lambda x_2 + x_3 &= 0 \\ x_2 - \lambda x_3 + x_4 &= 0 \\ x_3 - \lambda x_4 &= 0 \end{aligned} \tag{6}$$

if we set x_1 to 1,

$$\begin{aligned} &x_2 = \lambda \\ &x_3 = \lambda^2 - 1 = (\lambda - 1)(\lambda + 1) \\ &x_4 = \lambda - 1/\lambda \end{aligned} \tag{7}$$

note

$$\left[\frac{1}{2}(\pm 1 \pm \sqrt{5}) - 1\right]\left[\frac{1}{2}(\pm 1 \pm \sqrt{5}) + 1\right] = \frac{1}{2}(\mp 1 \mp \sqrt{5}) - 1 \tag{8}$$

$$\left[\frac{1}{2}(\pm 1 \mp \sqrt{5}) - 1\right]\left[\frac{1}{2}(\pm 1 \mp \sqrt{5}) + 1\right] = \frac{1}{2}(\mp 1 \pm \sqrt{5}) - 1 \tag{9}$$

$$\left[\frac{1}{2}(1 \pm \sqrt{5}) - 1\right] - \frac{1}{\left[\frac{1}{2}(1 \pm \sqrt{5}) - 1\right]} = 1 \tag{10}$$

$$\left[\frac{1}{2}(-1 \pm \sqrt{5}) - 1\right] - \frac{1}{\left[\frac{1}{2}(-1 \pm \sqrt{5}) - 1\right]} = -1 \tag{11}$$

for λ_1,

$$
\begin{aligned}
v_1 &= \left(1, -\frac{1}{2} - \frac{\sqrt{5}}{2}, \frac{1}{2} + \frac{\sqrt{5}}{2}, -1\right) \\
v_2 &= \left(1, \frac{1}{2} - \frac{\sqrt{5}}{2}, -\frac{1}{2} + \frac{\sqrt{5}}{2}, 1\right) \\
v_3 &= \left(1, -\frac{1}{2} + \frac{\sqrt{5}}{2}, \frac{1}{2} - \frac{\sqrt{5}}{2}, -1\right) \\
v_3 &= \left(1, \frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{1}{2} - \frac{\sqrt{5}}{2}, 1\right)
\end{aligned} \tag{12}
$$

2.

$$B = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix} \tag{13}$$

$$B - \lambda I = \begin{pmatrix} 1-\lambda & -1 & 0 & 0 \\ -1 & 2-\lambda & -1 & 0 \\ 0 & -1 & 2-\lambda & -1 \\ 0 & 0 & -1 & 1-\lambda \end{pmatrix} \tag{14}$$

the characteristic polynomial of B is

$$(1-\lambda)((2-\lambda)((2-\lambda)(1-\lambda)-1)-(1-\lambda))+((-(2-\lambda)(1-\lambda))+1) \tag{15}$$

which is equal to $x(x-2)(x^2-4x+2)$ which gives

$$\begin{aligned} \lambda_0 &= 0 \\ \lambda_1 &= 2-\sqrt{2} \\ \lambda_2 &= 2 \\ \lambda_3 &= 2+\sqrt{2} \end{aligned} \tag{16}$$

for eigenvectors, we know $v_0 = (1,1,1,1)$
we can also guess $v_2 = (1,-1,-1,1)$ and it verifies to be correct.
the final two we can also guess, the diagonal is $1 \pm \sqrt{2}, \pm\sqrt{2}, \pm\sqrt{2}, 1 \pm \sqrt{2}$

$$v_1 = (-1, 1-\sqrt{2}, -1+\sqrt{2}, 1) \tag{17}$$

$$v_3 = (-1, 1+\sqrt{2}, -1-\sqrt{2}, 1) \tag{18}$$

3.

$$C = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix} \tag{19}$$

one can guess the main eigenvector $(1,1,1,1)$ with eigenvalue 3.